# Software Test Plan (STP) - Smart Expenses and Budget Management Application

**Project:** Smart Expense & Budget Management Application

**Version:** 1.0

**Authors:** Viren James Fernandes (PES2UG24CS590), YV Prateek (PES2UG24CS617), Vikranth Acharyya (PES2UG24CS587), Vidushi Singh (PES2UG24CS581)

**Date:** 16/09/2026

**Team No:** 3

**Status:** Draft

## 1. Introduction

**Purpose:** This document defines the test plan for the Smart Expense & Budget Management Application v1.0. It outlines the objectives, scope, strategy, resources, schedule, and responsibilities for testing the system against the requirements defined in the Software Requirements Specification (SRS v1.0) and the design defined in the Software Architecture and Design specification (SAD v1.0).

**Scope:** Testing covers user registration and authentication, profile management, income and expense transaction management, categorization, search and filtering, budget creation and monitoring, savings goals, recurring transactions, financial summaries and graphical reports, report export, and budget-related notifications. Direct banking transactions, loan processing, investment management, and financial transactions with external banks are excluded from scope, consistent with SRS Section 2.5. Bank-account synchronization, AI-assisted category suggestions, multi-currency support, and shared/family budgets are listed as open issues/future work in the SAD and are therefore not tested in this release.

**References:**

- Software Requirements Specification (SRS) v1.0 – Smart Expense & Budget Management Application
- Software Architecture and Design (SAD) v1.0 – Smart Expense & Budget Management Application
- Requirements Traceability Matrix (RTM), SRS Section 8
- OWASP Application Security Verification Standard (ASVS)
- NIST SP 800-63B (Digital Identity Guidelines)
- WCAG 2.1 AA (Web Content Accessibility Guidelines)

**Definitions:** SRS (Software Requirements Specification), SAD (Software Architecture and Design), STP (Software Test Plan), UI (User Interface), API (Application Programming Interface), DB (Database), OTP (One-Time Password), CRUD (Create, Read, Update, Delete), NFR (Non-Functional Requirement), FR (Functional Requirement), RTM (Requirements Traceability Matrix), JWT (JSON Web Token), TLS (Transport Layer Security), WCAG (Web Content Accessibility Guidelines).

## 2. Test Items

- Authentication & Session Service (registration, login, password reset, profile updates, session/token management)
- Income & Expense Transaction Service (add, edit, delete, categorize, search/filter)
- Budget Management & Monitoring Service (budget creation, spend tracking, remaining budget/available balance)
- Savings Goal Service (create/manage goals, progress tracking)
- Recurring Transaction Scheduler (automatic generation of recurring income/expense entries)
- Summary & Reporting Service (daily/weekly/monthly summaries, graphical reports, export)
- Notification & Alert Service (budget threshold and limit alerts)
- Administrator interface (user account management, category management, system activity monitoring)
- Database module (Users, Transactions, Budgets, Categories, SavingsGoals, RecurringRules, Notifications)

## 3. Features to be Tested

| Req ID | Feature / Requirement | Type |
|---|---|---|
| SEA-F-001–004 | User registration, login, password reset, and profile management | Functional |
| SEA-F-005, 006 | Add, edit, and delete income and expense transactions | Functional |
| SEA-F-007 | Categorize income and expense transactions | Functional |
| SEA-F-008, 009, 010 | Create/manage budgets, track spending, and view remaining budget and available balance | Functional |
| SEA-F-011, 012 | Generate daily/weekly/monthly summaries and graphical spending reports | Functional |
| SEA-F-013 | Search and filter transactions | Functional |
| SEA-F-014 | Create and manage savings goals | Functional |
| SEA-F-015 | Generate budget threshold/limit alerts | Functional |
| SEA-F-016 | Manage recurring income and expenses | Functional |
| SEA-F-017 | Export expense and budget reports | Functional |
| SEA-SR-001–005 | HTTPS/TLS, password hashing, cross-user authorization, input validation, and session expiry | Security |
| SEA-NF-001 | Provide responsive performance for common operations | Non-Functional |
| SEA-NF-002 | Maintain system availability of 99.9% monthly | Non-Functional |
| SEA-NF-003 | Maintain data consistency for transactions, budgets, and savings goals | Non-Functional |
| SEA-NF-004 | Ensure accessibility according to WCAG 2.1 AA | Non-Functional |
| SEA-NF-005 | Maintain a modular, maintainable architecture, verified through integration and regression testing | Non-Functional |

## 4. Features Not to be Tested

- Direct banking transactions and access to users' bank accounts
- Loan processing
- Investment management
- Financial transactions with external banks
- Bank-account synchronization via an authorized banking API (SAD Section 4.6, out of scope for this release)
- AI-assisted category suggestions (SAD Section 4.6, future enhancement)
- Multi-currency support (SAD Section 4.6, future enhancement)

- Shared/family budgets with multiple contributing users (SAD Section 4.6, future enhancement)

## 5. Test Approach / Strategy

**Levels:**

- Unit tests (module-level, e.g., budget calculation, hashing routines)
- Integration tests (service-to-service and service-to-database, e.g., Transaction Service to Budget Service)
- System tests (end-to-end application functionality)
- Acceptance tests (UAT)

**Types:**

- Functional testing (core features per Section 3)
- Regression testing (after each build/fix)
- Performance testing (response time, load per SEA-NF-001)
- Security testing (authentication, authorization, input validation)
- Usability and accessibility testing (UI clarity, WCAG 2.1 AA conformance per SEA-NF-004)
- Compatibility testing (desktop, tablet, smartphone, supported browsers)

**Entry Criteria:** Stable build delivered, test data available, test environment ready, SRS/SAD baselined.
**Exit Criteria:** 100% of planned test cases executed, 0 critical/high-severity defects open, all acceptance criteria in SRS Section 6.3 satisfied.

### 5.1 Security Validation

- Verify HTTPS/TLS 1.2+ is enforced for all client-server communication (SEA-SR-001)
- Verify passwords are stored using a one-way hash and never in plaintext (SEA-SR-002)
- Verify cross-user access to another user's transactions, budgets, or savings goals is denied (SEA-SR-003)
- Server-side input validation / fuzzing of API request fields to check for injection and malformed-request handling (SEA-SR-004)

- Verify inactive authenticated sessions expire and cannot access protected screens (SEA-SR-005)
- Rate-limit testing on authentication endpoints (per SAD threat model — Denial of Service)

## 6. Test Environment

**Hardware:** Desktop/laptop, tablet, and smartphone devices with standard keyboard/mouse or touchscreen input.

**Software:** Smart Expense & Budget Management Application (web/mobile client), REST application server, relational database (PostgreSQL/MySQL) test instance, supported browsers.

**Tools:** Selenium (UI automation), Postman/Swagger (API testing), JMeter (performance/load testing), Jira (defect tracking), OWASP ZAP (security scanning), axe/WAVE (accessibility auditing).

**Test Data:** Dummy user accounts, sample transactions, budgets, categories, and savings goals covering normal, boundary, and error cases.

## 7. Test Schedule

| Milestone | Date |
|---|---|
| Test plan finalized | 16-Sep-2026 |
| Test case design | 18-Sep-2026 |
| Test environment setup | 19-Sep-2026 |
| Test execution start | 22-Sep-2026 |
| Test execution end | 03-Oct-2026 |
| UAT | 06-Oct-2026 to 09-Oct-2026 |
| Test summary report | 10-Oct-2026 |

## 8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual & automated)
- Test Scripts
- Test Data
- Test Execution Logs
- Defect Reports
- Requirements Traceability Matrix (updated with test status)

- Test Summary Report

## 9. Roles and Responsibilities

| Role | Name | Responsibility |
|---|---|---|
| QA Lead | Viren James Fernandes | Prepare test plan and coordinate test execution |
| Test Engineer | YV Prateek | Design and execute test cases; log defects |
| Test Engineer | Vikranth Acharyya | Design and execute test cases; log defects |
| Developer / Support | Vidushi Singh | Support defect fixes and triage |
| Course Coordinator / Product Owner | Prof. Pavithra M | Approve test results and sign-off readiness |

## 10. Risks and Mitigation

| Risk | Mitigation |
|---|---|
| Inconsistent budget totals under concurrent transaction updates | Test database transactions/row-level locking on budget updates; include concurrency test cases |
| Missed or duplicate recurring transactions | Test the idempotent scheduler and execution log; include boundary/edge-date test cases |
| Unauthorized cross-user data access | Dedicated authorization test suite covering every transaction, budget, and goal endpoint (SEA-SR-003) |
| Notification delivery failure | Test retry-queue and in-app alert fallback paths |
| Delay in stable build delivery | Request early smoke builds from the development team |
| Test environment downtime | Maintain a backup test environment on a cloud VM |

## 11. Assumptions & Dependencies

- The application server and database test environment will be stable and available
- Test data (accounts, transactions, budgets) will be provided or generated before execution begins
- SRS v1.0 and SAD v1.0 remain the approved baseline for the duration of testing

- Third-party notification/email gateway, if integrated, will provide a sandbox for testing

## 12. Suspension & Resumption Criteria

**Suspend testing if:**

- Test environment is unavailable for more than 4 hours
- Build is too unstable (blocks more than 30% of planned test cases)
- A critical security defect (e.g., cross-user data access) is discovered and remains unresolved

**Resume testing if:**

- Blocking defects are resolved
- Environment is stabilized and verified

## 13. Test Case Management & Traceability

The Requirements Traceability Matrix (RTM), maintained in SRS Section 8, ensures every functional, security, and non-functional requirement is mapped to one or more test cases. Examples:

- SEA-F-002 (Authenticate users) → TC-Auth-02
- SEA-F-009 (Track spending against budget limits) → TC-Budget-03
- SEA-F-015 (Budget alerts) → TC-Notify-01
- SEA-SR-002 (Password hashing) → TC-Sec-02
- SEA-NF-001 (Response time) → TC-Perf-01

The RTM status column will be updated from "N" (not tested) to "P" (passed) or "F" (failed) as execution progresses, and reviewed at each milestone in Section 7.

## 14. Test Metrics & Reporting

**Metrics collected:**

- % test cases executed
- % passed / failed
- Defect density and defect aging
- Requirement coverage (from the RTM)
- 90th-percentile response time against the 5-second target (SEA-NF-001)
- Monthly uptime against the 99.9% target (SEA-NF-002)

**Reports:**

- Daily execution status
- Final Test Summary Report

## 15. Approvals

| Role | Name | Signature / Date |
|---|---|---|
| QA Lead | Viren James Fernandes | |
| Test Lead | YV Prateek | |
| Dev Lead | Vikranth Acharyya | |
| Developer | Vidushi Singh | |
| Course Coordinator / Product Owner | Prof. Pavithra M | |